British Math Olympiad Round 1 2014 — P2/6

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Positive integers p, a and b satisfy the equation $p^2 + a^2 = b^2$. Prove that if p is a prime greater than 3, then a is a multiple of 12 and $2(p + a + 1)$ is a perfect square.

Solution

We begin by noticing that $p^2 = (b - a)(b + a)$. The two factors on the right are either p and p or they are 1 and p^2, but since $a > 0$ we must have $b + a = p^2$ and $b - a = 1$. Solve this simaltaenous equation to get $a = \frac{p^2-1}{2}$ and $b = \frac{p^2+1}{2}$. Notice that $2a = (p - 1)(p + 1)$, now since $p > 3$ and p is a prime 3 isn't a factor of p. But among the collection of numbers $p - 1, p, p + 1$ there must be one factor of 3, and thus one of $p - 1, p + 1$ is divisible by 3 and this means that a also has a factor 3. Similarly notice that p can't have a factor of 4 since it is odd, but among the collection of numbers $p - 1, p, p + 1, p + 2$ one of them must be a factor of 4. But since p and $p + 2$ are both odd they can't be divisible by 4, and thus one of $p - 1, p + 1$ is divisible by 4, and the other is divisible by 2 (since they are both even numbers), but this means that a has a factor of 4. Combining these 2 facts together shows that a is indeed a multiple of 12. Finally to show that $2(p + a + 1)$ is a perfect square observe the following

$$\begin{aligned} 2(p + a + 1) &= 2\left(p + \frac{p^2 - 1}{2} + 1\right) \\ &= p^2 + 2p + 1 \\ &= (p + 1)^2 \end{aligned}$$

and this completes the proof. ■